Candidate's name/ Index number ……

THE UNITED REPUBLIC OF TANZANIA
THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT
ANNUAL JOINT EXAMINATION

FORM ONE
032 CHEMISTRY
MARKING SCHEME NOVEMBER 2024

SECTION A (15 Marks)

1.

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
D	A	B	C	B	D	C	B	A	B

(01@ = 10 marks)

2.

List A	(i)	(ii)	(iii)	(iv)	(v)
List B	E	H	A	C	D

(01@ = 05 marks)

SECTION B (70 Marks)

3. (a)
(i) First aid is the help given to someone who is injured or sick before the victim gets further medical assistance. 02 marks
(ii) First aid kit is a small box containing materials for rendering first aid. 02 marks

(b) Importance of first aid
Relieve pain
Bring hope to the victim.
Prevent permanent disability
Prevent the victim's condition from getting worse
Reduce the possibility of death Any four 1½@ = 06 marks
Shorten recovery time

4.

- Production of heat for heating substances in the laboratory and also cooking: In this case, a non-luminous flame, which produces much heat, is more convenient.
- Flame tests for elements: When some elements are strongly heated, they produce characteristic flame colours that distinguish them from one another.
- Production of light: Flames produce light that can be used to light a dark room. Examples of heat sources, which produce flames that may be used for lighting, are hurricane lamp, tin lamp, spirit lamp and candle. 02½ @ = 10 marks
- Welding: In most welding operations, an oxyacetylene gas, a mixture of oxygen and ethyne, is used. When burned, the gas produces a flame hot enough to cut or melt the metal.

5. (a) An element is a substance with contains the same type of atoms/An element is a single pure substance that cannot be broken down to anything simpler by a chemical change. 02 marks

(b) (i) The assigning of symbols for chemical elements followed the following rules:

- The symbol is the capitalized initial letter of the element. For example, the symbol for hydrogen is H and Potassium (Kalium) is K. 02 marks
- The symbol is the first capital letter plus the second lower case letter of the element. For example, the symbol for calcium is Ca and for sodium (Natrium) is Na. 02 marks
- The symbol is the first capital letter plus any other lower case letter of the element. For example, the symbol for Zinc is Zn and Copper (Cuprim) is Cu. 02 marks

(ii)

Name	Symbol
Lithium	Li 00½ mark
Sodium	Na 00½ mark
Potassium	K 00½ mark
Magnesium	Mg 00½ mark

6. (a) A fire will continue or start to burn if these components are present:

(i) **Fuel:** This refers to any combustible material be it solid, liquid or gaseous material provided it can catch fire and burn. You can stop fire by removing the combustible material from the path of fire. 02 Marks

(ii) **Oxygen (air):** Oxygen supports combustion. A fuel will only burn if there is sufficient supply of oxygen. You can extinguish fire by displacing, or taking away oxygen supply from the fire or by blocking the gas supply to the fire. 02 Marks

(iii) **Heat:** For a fire to start burning, the temperature should be at the kindling point of that fuel or above it. Every fuel has its own kindling or ignition point. Below the ignition point, the fuel will not catch fire. You can put out fire by lowering the temperature below the kindling point of a particular fuel. 02 Marks

(b)<u>Similarity and differences between rusting and burning</u>

Similarity

- Chemically, rusting and burning are similar processes in that they both require oxygen.
- In addition, the two processes, burning and rusting, are exactly similar in that they both generate heat. 02 Marks

Differences: 02 Marks

- Burning produces much noticeable heat and light.
- Burning much rapid reaction in comparison to rusting.

7.

	Mixture (sea water)	**Compound (Pure water)**
1	The components of sea water (a mixture) can be separated by physical means, example, by distillation	The components of a compound can be separated by chemical means only.
2	The composition of a mixture can vary widely, example, sea water can contain various amount of salt and pure water.	Compounds are fixed in their compositions by mass of elements present, example, there

		are always 2 atoms of hydrogen to 1 atom of oxygen in a molecule of water
3	Formation of mixture is not usually accompanied by heat changes.	Formation of a compound is usually accompanied by heat changes.
4	Properties of a mixture are the sum of the properties of the individual constituents of the mixture.	The properties of a compound are entirely different from those of its constituent elements. For example, water is a liquid whereas its constituent elements, hydrogen and oxygen, are both gases.

$02½@ = 10$ marks

8. Life without the ability of matter to change states/ The importance of changing one state of matter to another

(i) Separation of mixture would have been impossible/it enables separation of mixtures can through such processes as distillation, sublimation, evaporation and condensation.

(ii) Industrial manufacture of products would have been hindered: Industrially, the process of distillation is applied in the production of pure substances such as beer and other alcoholic drinks such as wine. The manufacturing process involves boiling, evaporating and condensation.

(iii) Refining of crude oil would have been impossible: Crude oil contains organic liquid components, each with a different boiling point. In the refinery, the components with lower boiling points evaporate first and get separated out, leaving those with higher boiling points behind. In this way, we get various types of oil components (fractions) such as petrol, diesel, kerosene, lubricating oil, etc.

(iv) Drying would not be possible: When you suspend your clothing on a cloth line to dry, the moisture in it is lost through evaporation. Likewise, farmers in the village often spread crops on the ground to dry.

(v) Cooling of our bodies in hot weather depends in the evaporation of sweat or water.

(vi) You can freeze water into ice and then use the resulting ice for cooling the beverage or preserving foods.

(vii) In metallurgical industries, need may arise to mix two or more metals (alloys) together. This is only possible, where two or more metals are first melted at high temperatures into liquids.

(viii) There wouldn't rain: Formation of rain: Rain is mainly formed through the process of evaporation and condensation. Water vapour, evaporating mostly from water bodies (oceans, seas, lakes, rivers, ponds, etc), land and plants rises up to the sky. As it rises, it cools down and condenses into tiny droplets (Any five 02 @ = 10 marks)

9.

(a) - Doors open outward for easy departure during emergency and for reducing the chance of knocking down laboratory apparatus and equipment.

- The fume chambers are designed to capture and remove hazardous airborne substances generated during laboratory experiments that produce hazardous waste

products are performed.

(b)

Solutions	Suspensions
…homogeneous	...heterogeneous
Transparent	Opaque
Particles completely dissolved	Particle separate on standing
Components separated by evaporation	Components separated by filtration

(Any four 1½ @ = 06 marks)

SECTION C (15 MARKS)

10. The scientific method (procedure) is a process that scientists use to conduct investigations to find answers to problems. It is a logical approach to problem solving by formulating hypotheses, observing and collecting data, testing hypotheses, and formulating theories that are supported by data. 03 marks

<u>The Importance of the Scientific Procedure</u>

i. The scientific procedure makes a researcher or an experimenter more systematic and organized when investigating or solving a problem.
ii. It gives a means by which one can get a solution to several questions about natural phenomena.
iii. It may lead to discoveries and innovations. Any six 02 @ = 12 marks
iv. Provides background knowledge upon which future references may be made.
v. It makes our sense organs more effective in exploring our natural world. That is, we become more sensitive to environmental changes.
vi. It makes us use the available resources more sustainably in solving everyday problems.
vii. Assists us in predicting the future outcome based on the present condition.
viii. Assists us in testing the validity or the possibility of an event, phenomenon or problem.